AI Lab: Comprehensive RAG Challenge

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***Abstract*—In this AI Lab project, we reimplemented and extended a retrieval-augmented generation (RAG) system originally proposed by Team Apex for the CRAG (Comprehensive RAG) benchmark. Our work focused on adapting the pipeline to local computing resources, adjusting prompt designs, and integrating newly introduced Chain-of-Thought (CoT) techniques. After replacing the original LLM with Llama3.3 70B and introducing few-shot prompts, we observed consistent performance improvements, albeit with a trade-off between accuracy and hallucination. Further refinements involved referencing source documents directly in the CoT examples, which boosted domain-specific reliability (especially in music-related queries). We also highlight some challenges, such as mismatched ground truths in the CRAG dataset and time-consuming runs on large-scale clusters. Our results suggest that larger-parameter models, domain-specific prompt strategies, and carefully curated references can significantly improve RAG systems in both accuracy and groundedness.**

I. RAG Challenge Overview

Large Language Models (LLMs) can excel at many NLP tasks but often struggle to remain factually accurate in fast-changing or specialized contexts. *Retrieval-Augmented Generation (RAG)* addresses this by fusing LLMs with external knowledge sources, aiming for more reliable and current information. However, RAG systems can face issues such as domain adaptation, dynamic data needs, and large volumes of retrieved documents.

To evaluate such systems, the **CRAG (Comprehensive RAG) Benchmark** [1] presents 4,409 question-answer pairs from five domains (finance, sports, music, movie, and open), each featuring time constraints, extensive web data, and structured Mock APIs. In this setting, the Apex team proposed a *routing-based* and *dynamic adaptive* RAG system, ultimately securing second place on Tasks 2 and 3.

II. Team Apex's Implementation

A. High-Level Pipeline

The Apex pipeline (Figure 1) combines unstructured web pages with domain-specific APIs to handle domain variation and quickly changing facts.

Domain Router. Classifies each query into finance, sports, music, movie, or open. This guides the system to call relevant APIs and apply the right prompt templates.

Dynamism Router. Determines how static or time-sensitive the question is (e.g., fast-changing vs. static). If no up-to-date data can be retrieved, the system returns "I don't know" rather than guess.

Retrieval.

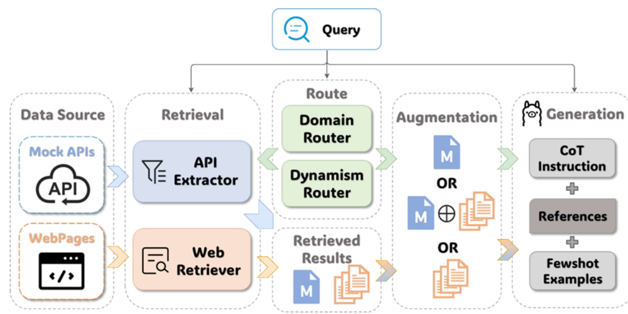

Fig. 1. Apex team's RAG pipeline.

- *Web Pages:* HTML is parsed, then divided into chunks. BM25 ranking filters out noisy segments, and a dense re-ranker (e.g., bge-m3) picks the most relevant chunks.
- *APIs:* Queries are processed for named entities (e.g., ticker symbols) and time references (e.g., "yesterday" mapped to a date). Simple rules pick the right domain-specific API, which returns JSON reformatted for large language model input.

Generation. All retrieved chunks (web + API) are merged into a single context for the LLM. *Chain-of-Thought (CoT)* prompting helps the model reason explicitly, and few-shot examples guide it to refuse speculation when data is missing or out of date.

B. Evaluation Results

Table I shows how Apex compares to both a baseline LLM and a simple RAG approach. The improved accuracy and reduced hallucinations reflect the benefit of domain-aware routing and dynamic adaptation. The trade-off is a higher missing rate, as the system avoids guessing when it lacks reliable information.

TABLE I
Evaluation of Apex on the CRAG benchmark.

System	Score	Accuracy (%)	Halluc. (%)	Missing (%)
LLM Only	-7.29	28.01	35.30	36.69
Direct RAG	-6.78	34.79	41.58	23.63
Apex (Task 1)	11.82	29.98	18.16	51.86
Apex (Task 2)	31.22	46.75	15.54	37.71

III. Our Works

A. Re-Implementation of team Apex's work

When we began this project, our initial goal in the lab was to understand Team Apex's implementation [2] and the state-of-the-art LLM pipeline using RAG. We started by downloading and processing the baseline provided by META for the competition, which was a prerequisite for all participating teams. Next, we installed the required packages and ran Team Apex's pipeline, making minor modifications to ensure compatibility with the L3S infrastructure that was provided for this lab. Through this process, we encountered and addressed several challenges:

- **Neumann:** We learned to use the GPU cluster Neumann, managed by the SLURM job scheduling system. This involved running Neumann from our local computer using VSCode and submitting jobs to execute the pipeline. Neumann offers various LLMs, allowing smooth runs of large-scale predictions.
- **Required Packages:** Because the exact versions of the necessary packages were not specified by the team, we had to experiment with multiple configurations to identify the correct versions. Some packages conflicted with one another, making version alignment crucial for a stable pipeline.
- **Router Weights:** The weights for the Domain Router and Dynamism Router were initially unavailable, leading to discrepancies when we reran the team's implementation. Once we obtained these weights by request, we were able to reproduce the results reported in their paper.
- **Interweb:** We attempted to run the LLMs through the Interweb resource but occasionally ran into repeated timeout issues due to the limited number of tokens given per user. Generation of predictions for the full dataset of 2,706 questions can take up to 10 hours, so we had to wait a considerable amount of time between evaluations whenever we updated the model.

Our re-implementaion could be found on GitHub.

B. Adjustment of Prompts

To fully understand how to improve the prompts for the prediction model, we modified the model (class *RAGModel* in *model.py*) so that it prints its thought process every time a question is processed. Based on the model's thought process, we were able to identify ways to refine the prompts.

1) General Note: We experimented with adding more general instructions for the prediction model. These guidelines are designed to encourage the model to answer questions concisely and directly, without overthinking. This adjustment was necessary because the model was taking too long and generating an excessive amount of thought for many questions in the provided dataset. The added notes are:

- If the query is a question, process it as instructed.
- If the query is missing or unclear, assume the user is asking a factual question.
- Avoid reinterpreting the input; proceed directly with answering.
- Avoid overthinking the queries and keep the answer concise.

These guidelines ensure that the model does not waste time by providing answers to questions it cannot answer, offering false answers, or classifying them as invalid.

2) Fewshot Chain-Of-Thought Prompts: The original prompt templates developed by Team Apex to guide the LLM in answering questions across different domains do not differ significantly, as they share a common structure and only vary in the examples of invalid questions provided. Additionally, the prompts did not include any Chain-Of-Thought (CoT) examples to guide the model beyond the standard instruction to "think step by step."

Based on the benefits of including few-shot prompts for enhancing the model's ability to answer domain-specific questions [3], we designed examples for the music, sport, finance, and movie domains (no examples were designed for the open domain due to its greater variety of questions). The examples feature question types similar to those in the dataset; however, we selected different examples with answers sourced externally to prevent the model from overfitting on them. We ensured that these examples did not appear in the dataset prior to their use.

Prompt generation can also be performed by AI models themselves, since they are capable of writing prompts autonomously [4]. We leveraged this capability to generate concise, question-specific instructions that guide the model more effectively toward the final answer. For this purpose, we used GPT-O1 from OpenAI, a recently introduced model with advanced reasoning capabilities that achieves human-level or superior performance in tasks ranging from coding challenges to scientific reasoning [5], making it well-suited to our requirements.

Initially, we tested with 8 examples per domain, as it was suggested that the LLM would achieve optimal and converged performance across various benchmark datasets [3]. However, using 8 examples caused the model to exceed its token limit, resulting in errors in our predictions. We therefore reduced the number of examples to 4 per domain, since the COT paper demonstrated that for domain-specific tasks (e.g., sports) the performance with 4 examples was comparable to that with 8. With this amount of examples per domain, we managed to run the model smoothly through all the questions in the dataset.

Some CoT examples for the movie domain are shown below:

Chain-of-Thought Examples for the Movie Domain

1) **Question:** What movies has Leonardo DiCaprio won an Oscar for?

   Thought: Identify all movies featuring Leonardo DiCaprio, filter for Oscar wins, confirm using official records.

   Final Answer: The Revenant.
2) **Question:** How many animated movies has Pixar released?

   Thought: Identify Pixar's complete filmography, count animated movies, exclude non-animated works.

   Final Answer: Total number of movies.
3) **Question:** What is the runtime of ”The Godfather Part II”?

   Thought: Verify the official runtime of ”The Godfather Part II” from reliable sources.

   Final Answer: 202 minutes.
4) **Question:** What was the highest-grossing movie in 2020?

   Thought: Review global box office data for 2020, confirm the movie with the highest revenue.

   Final Answer: Demon Slayer: Mugen Train.

The results of running Task 2 with CoT are presented in table II.

3) Addition of References: Noticing comparatively lower performance in the music domain, we explored a different prompt-engineering strategy for better results. Inspired by *Retrieval Augmented Fine-Tuning (RAFT)*, a 2024 approach that fine-tunes models on QA data with an explicit chain of thought citing supporting documents [6], we adapted a similar technique. In RAFT, each training sample contains the question, relevant oracle document(s), plus step-by-step reasoning that quotes the oracle content and culminates in the final answer. Models trained using this reference-grounded CoT format (“Question $\rightarrow$ References $\rightarrow$ Thought $\rightarrow$ Answer”) significantly outperform those trained on standard QA or CoT alone.

Following RAFT's principles, we designed new CoT examples that include an additional “References” line. To prevent token limit issues, we limited our revised CoT examples to two instances for the music domain, since each example now includes extra lines for references. The revised CoT examples appear below:

Chain-of-Thought Examples for the Movie Domain

1) **Question:** What songs did Whitney Houston release in the 1990s?

   References: ”Whitney Houston's 1990s albums include 'I'm Your Baby Tonight' (1990), 'The Bodyguard Soundtrack' (1992), 'Waiting to Exhale Soundtrack' (1995).”

   Thought: Summarize key releases from references.

   Final Answer: I'm Your Baby Tonight, The Bodyguard Soundtrack, Waiting to Exhale Soundtrack.
2) **Question:** Who won the Grammy for Record of the Year in 2010?

   References: ”The 2010 Grammy for Record of the Year went to 'Use Somebody' by Kings of Leon.”

   Thought: Direct info from references.

   Final Answer: Kings of Leon, Use Somebody.

IV. Results

The evaluation results for all domains combined, as well as each individual domain, are presented in Table II. All outcomes were obtained using Llama3.3-70B for our LLMs (whereas the original authors employed an earlier version, Llama3-70B).

TABLE II

Our evaluation results on the CRAG bench mark.

System	Score	Accuracy (%)	Halluc. (%)	Missing (%)
Task 1				
w/o Adjustment	16.81	30.75	13.93	55.32
Task 2, CoT v1				
Overall	32.07	49.74	17.66	32.59
Finance	34.64	43.12	8.47	48.41
Movie	33.06	60.56	27.50	11.94
Music	19.84	52.55	32.71	14.75
Sports	32.76	36.60	3.85	59.54
Open	35.61	56.27	20.66	23.06
Task 2, CoT v2				
Music	21.72	34.32	12.60	53.08

A. Task 1

For Task 1, we made no changes to the pipeline other than substituting Llama3.3-70B, yet still achieved a higher score and accuracy alongside a lower hallucination rate compared to the authors' results. This suggests that the newer version of Llama offers superior performance, consistent with other benchmark findings [7]. It follows that upgrading to a more advanced model can further enhance the pipeline's effectiveness.

B. Task 2

1) Fewshot CoT Prompts: For Task 2, we used newly generated Chain-of-Thought (CoT) examples and supplemented the prompts for our prediction model with additional notes. We observed improved Score, Accuracy, and Missing metrics overall. While Score and Accuracy increased slightly, hallucinations also rose as fewer questions were answered with “I don't know.” A likely explanation is that the modified prompts encouraged the model to generate answers more often, leading to more correct responses but also some incorrect ones when the necessary information was unavailable in the database. This is also mentioned in the research [8], where the authors observe that RAG models often face a tension between coverage and correctness: pushing the model to produce an answer even when retrieval is sparse can lead to factually inaccurate completions.

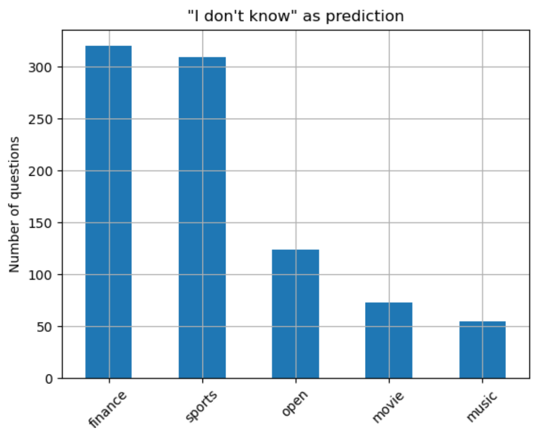

Fig. 2. Number of "I don't know" answers

Since we were not fully satisfied with evaluating our results only on the entire dataset—particularly because we wanted to see whether the models performed differently across domains—we split the predictions into five domain-specific files and evaluated them separately. As shown in Table II, all domains except for music exhibit similarly strong performance. Notably, the music domain has the fewest "I don't know" responses (Figure 2) but also demonstrates the highest hallucination rate among all domains. This pattern suggests that stricter prompts for music induced the model to generate more speculative answers rather than defaulting to "I don't know," thereby increasing hallucinations and reducing the final score. By contrast, the same prompt strategy in other domains still offered a favorable balance of correctness. These findings indicate that not every domain benefits equally from identical prompting techniques.

2) Addition of References: As we were dissatisfied with the model's performance in the music domain, we investigated and developed an alternative prompt-engineering approach to achieve better results. By adding dedicated "References" lines to the prompts for music-related queries, we observed an improvement in the domain's Score (see Table II, Task 2, CoT v2). Consequently, if the other domains maintain their current level of performance, the overall dataset Score could reach at least 32.34. This outcome indicates that the newly adopted technique can serve as a beneficial enhancement for RAG-based LLMs and may be extended to other domains as well.

V. Future Work

A. Scaling models

We mainly worked with Llama3.3 70B for both prediction and evaluation, while the authors used an earlier version (Llama3.1) with the same parameter size. As a result, we consistently achieved slightly better results than those reported in their work. We also experimented with a smaller-scale model, Gemma2 27B, for evaluation, which produced significantly lower scores than Llama3.370B (see Table III). Consistent with recent studies [3], using models with more parameters generally yields significantly better performance. We therefore recommend employing higher-parameter models whenever resources allow, to maximize results.

B. Adaptive strategies for question types

As we designed prompts only for different domains and did not tailor them to specific question types, there remains an opportunity to enhance the results. In addition to domain-based routing, we could also introduce question-type routers that redirect specific queries to prompts specifically designed for each question type. Given that the dataset includes eight question types, we could develop eight different sets of few-shot CoT examples per domain. While this would likely slow down prediction generation due to the increased complexity, it could yield promising results if sufficient computational resources are available.

C. Incorporating the HoT Technique.

A promising direction for future work is to adopt the newly introduced Highlighted Chain-of-Thought (HoT) approach [9], which explicitly highlights key evidence within the model's reasoning path. By visually marking crucial information, the model can better align its answers with the relevant context, potentially reducing hallucinations and enhancing factual consistency. Integrating HoT into our existing pipeline would likely strengthen result quality, especially in domains where context identification is critical.

D. Experimentation with Multiple Benchmark Datasets

While working with the CRAG dataset, we discovered that some question sets contained incorrect ground truths, as verified by our own external searches. This issue can negatively affect our evaluation scores, since our model may produce correct answers that are erroneously labeled due to faulty references. Although our results remain competitive in the context of the competition, for a more reliable assessment of the model's performance, we plan to experiment with other rigorously vetted benchmark datasets. One strong alternative is the *KILT* (Knowledge Intensive Language Tasks) benchmark [10], which satisfies key criteria such as realism, richness, insightfulness, reliability, and longevity. KILT is particularly promising for future RAG-based research, as it more comprehensively captures the multifaceted challenges of knowledge-intensive language processing.

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APPENDIX

TABLE III
EVALUATION RESULTS WITH GEMMA 2 27B

System	Score	Accuracy (%)	Halluc. (%)	Missing (%)
Task 1				
Gemma 2 27B	-17.85	13.41	31.26	55.32